

SplineLoc: Single-Scan Cross-View Localization from Rover LiDAR to Coarse Lunar DEMs

Anonymous ICRA Submission

Abstract—We present SplineLoc, a single-scan cross-view localization method that matches rover LiDAR against coarse lunar digital elevation models (DEMs) by representing both as continuous cubic B-spline height fields. Existing LiDAR global localization methods typically rely on reference-scan databases, while planetary approaches often require landmarks, temporal accumulation, or bounded initialization, making direct single-scan localization against coarse orbital DEMs challenging. SplineLoc addresses these gaps by representing both the DEM and a levelled LiDAR scan as continuous cubic B-splines, performing map-wide retrieval with masked Fast Fourier Transform (FFT) correlation, and refining the retained candidates directly on measured LiDAR points. Experiments on four lunar sites built from real orbital topography, including NASA LROC NAC, demonstrate that SplineLoc outperforms the state-of-the-art baselines across DEM resolutions up to 10 m/pixel, and maintains high accuracy as the reference lunar DEM becomes progressively coarser. Code will be available at: <https://anonymous1312602.github.io/SplineLoc-ICRA/>

I. INTRODUCTION

Future lunar exploration increasingly calls for long-range and autonomous surface mobility. The Artemis program and mission concepts such as Endurance-A, which envision highly autonomous rovers traversing nearly 2,000 km across the South Pole–Aitken basin, motivate reliable localization over extended traverses [1]. Without GNSS, rover odometry accumulates drift and requires periodic global corrections. LiDAR provides illumination-independent terrain geometry, while orbital digital elevation models (DEMs) provide a globally referenced prior over large areas. However, available lunar DEMs commonly have meter-scale sampling and can be coarser than rover LiDAR observations [2], [3].

Existing localization methods do not directly address this setting. General LiDAR global localization methods such as RING [4] and BEVPlace++ [5] typically retrieve against reference-scan databases, while planetary approaches often rely on extracted terrain features or accumulated observations [6]. Most closely related, Zhang et al. [7] match LiDAR measurements against a fitted DEM surface within Monte Carlo localization. However, they initialize particles inside a 20×20 m region, assuming a strong prior on the rover position, which does not address map-wide localization from an unknown initial position.

Prior work has studied cross-view localization in lunar and other planetary environments, but rover-to-orbital approaches predominantly use visual observations or extracted geometric primitives. Existing methods match surface imagery against orbital images [8], [9], [10], [11], or associate craters and horizon profiles with orbital maps [12], [13], [14]. Such approaches demonstrate the utility of orbital priors in natural

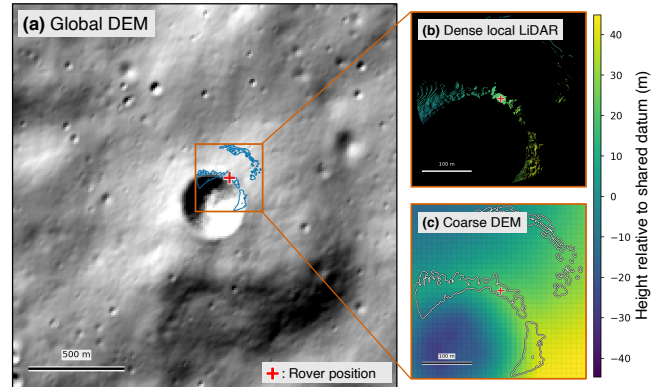


Fig. 1. **Dense local LiDAR versus a coarse global DEM.** (a) Global reference DEM with the rover's local observation region highlighted. (b) Dense LiDAR measurements within the region. (c) The corresponding coarse DEM patch at the same location. The large difference in sampling density motivates a common representation for LiDAR-to-DEM localization.

terrain, but rely on appearance or identifiable landmarks. A complementary line instead exploits lunar surface geometry itself: Huang et al. [15] constrain visual–LiDAR SLAM with locally planar patches. Such priors approximate the surface by a fixed local model, whereas SplineLoc encodes the same smoothness as a continuous spline height field that supports map-wide matching.

To address these limitations, we propose SplineLoc, a single-scan LiDAR-to-DEM localization method that requires neither position nor heading initialization, temporal accumulation, terrain landmarks, nor a reference-scan database. SplineLoc represents the levelled LiDAR scan and DEM as continuous cubic B-spline height fields and matches their relative terrain-height structure. We use masked Fast Fourier Transform (FFT)-based correlation for efficient global retrieval and refine the retained candidates directly on measured LiDAR points for metric pose estimation. We evaluate SplineLoc across multiple lunar terrains and DEM resolutions, where it consistently outperforms the evaluated baselines and maintains high localization accuracy as the reference DEM becomes progressively coarser.

Our contributions are:

- We employ a common cubic B-spline height-field representation for a coarse orbital DEM and a single LiDAR scan, supporting continuous evaluation and analytic gradients away from the source samples.
- We propose an offset-invariant height objective that decomposes into masked cross-correlations, making a map-wide search over translations and headings tractable by FFT, and that is reused as the refinement

cost on raw points.

- Evaluation against planetary and LiDAR localization baselines on four simulated lunar terrains and six DEM resolutions, a terrain-representation study and a reference-map experiment using native orbital products.

II. RELATED WORK

Planetary terrain-map localization. Carle et al. [6] matched topographic peaks extracted from rover LiDAR against an orbital elevation map, with multi-frame odometric integration improving robustness. Geromichalos et al. [16] periodically registered an accumulated local elevation map to an orbital map within particle-filter SLAM, and Le Gentil et al. [17] used Gaussian-process gradient maps for loop closure in planetary-analog terrain. Zhang et al. [7] incorporated a DEM-derived fitted surface into LiDAR-inertial SLAM through Monte Carlo localization initialized within a supplied region of interest. Recent systems adopt surface or map priors while tracking: Huang et al. [15] impose plane constraints within global optimization, and Dai et al. [18] anchor visual SLAM to a DEM. These methods depend on repeatable terrain features, temporal accumulation, or a bounded initialization region. In contrast, SplineLoc targets map-wide localization from a single LiDAR scan without terrain-landmark extraction, temporal accumulation, or a bounded position initialization.

Rover-to-orbital visual and landmark methods. Visual methods relate surface imagery to orbital imagery through geometric alignment [8] or learned lunar cross-view matching [9], [11]; Censible [10] localizes Mars panoramas against orbital imagery given an externally estimated orientation and an odometry-bounded search region. ShadowNav [12] associates observed craters with an orbital crater map, Ye et al. [14] match craters coarse-to-fine across multiple rover views and Tian et al. [13] match horizon profiles predicted from a DEM. These methods rely on visual appearance or on geometric primitives that must remain recoverable across the rover and orbital views.

LiDAR global localization. General LiDAR global localization commonly separates place retrieval from metric pose estimation. RING [4] uses a Radon sinogram, BEV-Place++ [5] learned rotation-equivariant BEV features, and CrossLoc3D [19] bridges aerial and ground LiDAR. These methods retrieve against a database of reference scans; SplineLoc instead evaluates candidate poses directly against a supplied DEM.

Continuous map representations. B-splines have been used as continuous representations in SLAM, including feature curves [20] and online scan-to-map surfaces [21]. We instead use a B-spline height-field representation to place a pre-existing orbital DEM and a single rover LiDAR scan in a common continuous form for global localization.

III. PROPOSED METHOD

A. Overview

SplineLoc localizes a single levelled LiDAR scan in a georeferenced DEM using three stages (Fig. 2). First, we

represent the DEM and LiDAR scan as continuous cubic B-spline height fields so that both observations can be compared in the same form (Sec. III-C). Second, we search the full map by comparing their relative terrain-height patterns over discrete positions and headings (Sec. III-D). Finally, we refine the retained hypotheses in continuous pose coordinates using measured LiDAR points and the analytic map gradient (Sec. III-E). Retrieval uses spline-derived rasters for efficient global search, whereas refinement uses the measured points directly for metric pose estimation.

B. Problem Statement

The inputs are a DEM, i.e., elevations on a georeferenced grid of spacing r_D with a validity mask, and a single LiDAR scan levelled to the map vertical using roll and pitch from an external attitude estimator. SplineLoc uses only roll and pitch for scan levelling and estimates the remaining planar pose through scan-map matching. Each LiDAR point is a pair (s_i, z_i) , where $s_i \in \mathbb{R}^2$ is its horizontal coordinate in the levelled sensor frame and z_i is its vertical coordinate relative to the sensor origin. The unknown pose is $\xi = [c_x, c_y, \psi]^\top$, comprising east-north position $\mathbf{c} = [c_x, c_y]^\top$ and heading ψ . Under a candidate pose, the horizontal map-frame position of point i is

$$\mathbf{p}_i(\xi) = \mathbf{R}_\psi \mathbf{s}_i + \mathbf{c}, \quad (1)$$

where $\mathbf{R}_\psi$ is the planar rotation matrix. We search the full extent of the supplied DEM without position or heading initialization and treat each scan as a rigid observation with negligible or compensated motion distortion.

C. Continuous Height-Field Representation

The DEM and LiDAR scan have very different sampling patterns, but both describe terrain height over the horizontal plane. We therefore express both observations as continuous height fields. The spline does not recover terrain details that are absent from the input; rather, it provides a common representation in which height and spatial gradient can be evaluated between the source samples.

Let β^3 denote the centred cubic B-spline kernel [22]. On a uniform knot grid of spacing h , the basis functions are $B_j(x) = \beta^3((x - x_j)/h)$, and the corresponding tensor-product height field is

$$f(x, y; \mathbf{C}) = \sum_{j,k} C_{jk} B_j(x) B_k(y), \quad (2)$$

where $\mathbf{C}$ is the coefficient grid. The compact support of the cubic basis limits each evaluation to 4×4 coefficients, while its C^2 continuity provides an analytic spatial gradient ∇f .

We use this representation for the map field f_M and scan field f_S . For the scattered LiDAR samples, we obtain the scan coefficients by penalized least squares [23],

$$\mathbf{C}_S = \arg \min_{\mathbf{C}} \sum_i [z_i - f(s_{ix}, s_{iy}; \mathbf{C})]^2 + \alpha_S \mathcal{R}(\mathbf{C}), \quad (3)$$

where $\mathcal{R}(\mathbf{C})$ is a second-difference smoothness penalty along both axes, suppressing poorly constrained local variation in sparsely sampled regions. For the regular-grid DEM,

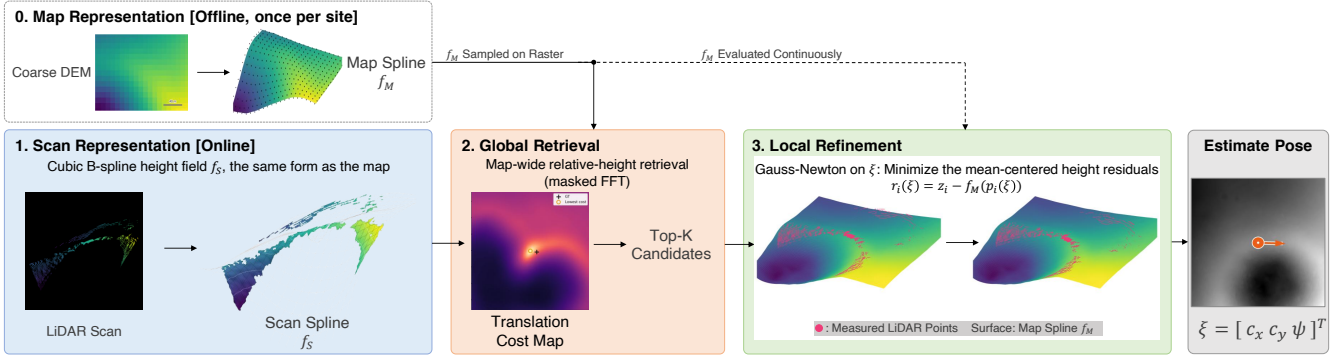


Fig. 2. **SplineLoc pipeline.** The coarse DEM and levelled LiDAR scan are first represented as continuous cubic B-spline height fields. We perform map-wide retrieval using an offset-invariant relative-height objective evaluated by masked FFT over translations and sampled headings, each candidate pose is then refined in continuous (c_x, c_y, ψ) by Gauss-Newton on the height residuals between the measured LiDAR points and the continuous map spline, and the K refined candidates are re-ranked by this point-based cost.

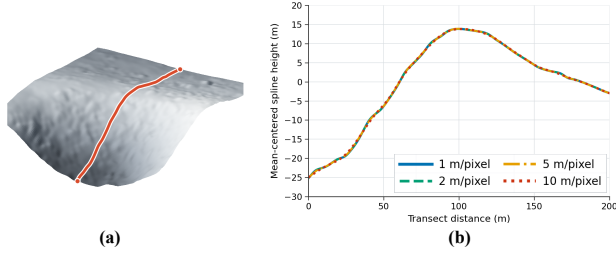


Fig. 3. **Map-spline profiles across DEM resolutions.** (a) Reference DEM and the selected terrain transect. (b) Fitted spline profiles along the same 200 m transect for DEMs ranging from 1 to 10 m/pixel. The spline representation preserves consistent broad terrain-height variations despite progressively coarser input DEMs.

we obtain C_M using a separable penalized fit that preserves the DEM grid structure [24].

The map knot spacing is $h_M = \max(h_0, r_D)$, avoiding a coefficient grid finer than the source DEM when $r_D > h_0$; otherwise the map is represented at the fixed scale h_0 . To examine what terrain information survives coarsening, we sample the fitted map spline along a fixed transect and compare its mean-centred height profile across DEM resolutions. Fig. 3 compares the fitted map fields along the same terrain transect as the DEM resolution is coarsened. Despite the reduced source sampling, the splines preserve nearly the same broad relative-height profile, with only small local deviations. This behavior motivates matching terrain structure at the represented scale rather than relying on fine-scale DEM detail. We fix the scan spacing h_S . For retrieval, we sample f_M and f_S on a common lattice, whereas refinement evaluates f_M and ∇f_M directly at transformed LiDAR points.

D. Global Retrieval

Relative-height matching. For a candidate heading and translation, SplineLoc places the query raster over the map and compares terrain heights only where the LiDAR-supported query overlaps valid DEM cells. Because the scan and DEM may differ by an unknown constant vertical offset, we remove the mean height difference over the overlap. A good candidate therefore matches the spatial pattern of terrain-height variation rather than the absolute elevation. Fig. 4 visualizes this retrieval process by comparing the

same LiDAR-derived query against DEM patches at different translations and sampled headings. Each pose hypothesis receives an offset-invariant relative-height cost, and evaluating these hypotheses over the full map yields the global shortlist used for refinement.

We discretize the continuous fields on a common two-dimensional lattice of spacing Δ . Let $\mathbf{u} = [u_x, u_y]^\top \in \mathbb{Z}^2$ be a lattice index and $\mathbf{o}_M \in \mathbb{R}^2$ the metric position of the map-lattice origin. Sampling the map spline gives

$$F(\mathbf{u}) = f_M(\mathbf{o}_M + \Delta\mathbf{u}),$$

where the binary mask $V(\mathbf{u}) \in \{0, 1\}$ indicates whether the corresponding map location is valid in the source DEM; we set $F(\mathbf{u}) = 0$ wherever $V(\mathbf{u}) = 0$.

For each sampled heading ψ , we construct a sensor-centred query raster on the same lattice. Index $\mathbf{u}$ corresponds to the scan location $\Delta\mathbf{R}_\psi^\top \mathbf{u}$. We define the support mask $m_\psi(\mathbf{u})$ to be one when this location lies within horizontal distance ρ of at least one measured LiDAR point and zero otherwise. The query raster is

$$q_\psi(\mathbf{u}) = m_\psi(\mathbf{u}) f_S(\Delta\mathbf{R}_\psi^\top \mathbf{u}), \quad (4)$$

so m_ψ identifies scan-supported cells and q_ψ stores their spline heights.

For a fixed heading ψ , let $\boldsymbol{\tau} \in \mathbb{Z}^2$ denote a candidate translation. Query cell $\mathbf{u}$ is then compared with map cell $\mathbf{u} + \boldsymbol{\tau}$. The number of supported cells in valid map overlap is

$$n_\psi(\boldsymbol{\tau}) = \sum_{\mathbf{u}} m_\psi(\mathbf{u}) V(\mathbf{u} + \boldsymbol{\tau}). \quad (5)$$

Define the height difference and its overlap mean as

$$d_{\psi, \boldsymbol{\tau}}(\mathbf{u}) = q_\psi(\mathbf{u}) - F(\mathbf{u} + \boldsymbol{\tau}),$$

$$\bar{d}_{\psi, \boldsymbol{\tau}} = \frac{1}{n_\psi(\boldsymbol{\tau})} \sum_{\mathbf{u}} m_\psi(\mathbf{u}) V(\mathbf{u} + \boldsymbol{\tau}) d_{\psi, \boldsymbol{\tau}}(\mathbf{u}). \quad (6)$$

We score the candidate by the mean-centred squared height residual,

$$E_{\text{ret}}^2(\boldsymbol{\tau}, \psi) = \frac{1}{n_\psi(\boldsymbol{\tau})} \sum_{\mathbf{u}} m_\psi(\mathbf{u}) V(\mathbf{u} + \boldsymbol{\tau}) \cdot [d_{\psi, \boldsymbol{\tau}}(\mathbf{u}) - \bar{d}_{\psi, \boldsymbol{\tau}}]^2. \quad (7)$$

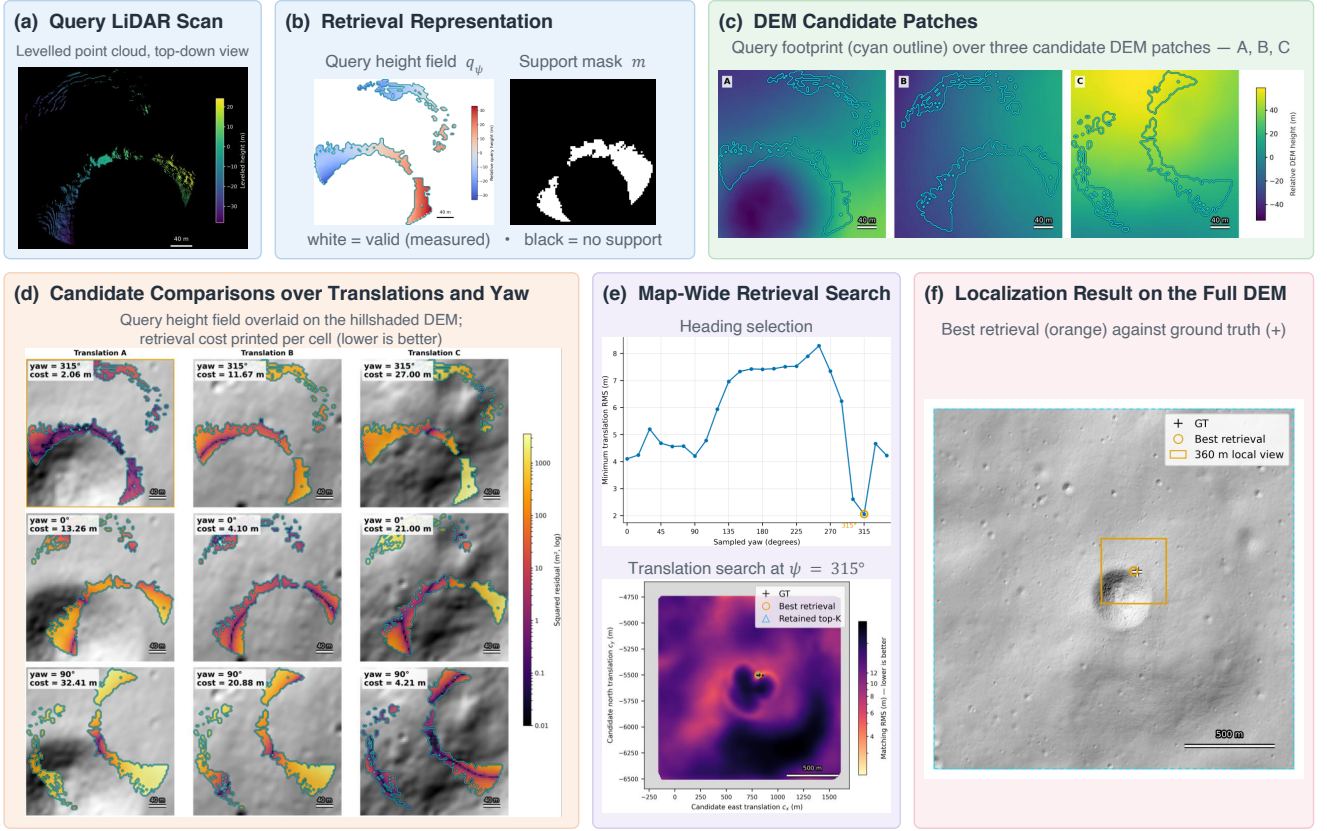


Fig. 4. **Map-wide retrieval over translations and headings.** (a–b) We form a supported relative-height query from the levelled LiDAR scan. (c–d) We compare this query with DEM patches at different map translations and sampled headings, assigning each pose hypothesis an offset-invariant height-matching cost. Lower costs indicate stronger agreement in relative terrain-height structure. (e) We evaluate the same objective across the full map and all sampled headings, then retain the top- K hypotheses for continuous refinement.

This is equivalent to minimizing the squared height discrepancy over an unknown constant vertical offset b . Equation (7) therefore compares relative terrain-height variation rather than requiring the scan and DEM to share the same absolute vertical level.

Efficient evaluation by masked FFT. Naively evaluating Eq. (7) at every translation would repeatedly sum over the query support. The key observation is that translation changes only the relative shift between the query and map rasters. After expanding the mean-centred squared error, every translation-dependent term becomes a discrete cross-correlation. With

$$(A \star B)(\tau) = \sum_{\mathbf{u}} A(\mathbf{u})B(\mathbf{u} + \tau),$$

we have $n_\psi = m_\psi \star V$, and

$$E_{\text{ret}}^2(\tau, \psi) = \frac{(q_\psi^2 \star V) - 2(q_\psi \star F) + (m_\psi \star F^2)}{n_\psi} - \left[\frac{(q_\psi \star V) - (m_\psi \star F)}{n_\psi} \right]^2, \quad (8)$$

where every correlation and n_ψ is evaluated at τ . Thus, for each sampled heading, the complete translation-cost map is obtained from a small set of correlations rather than by scoring translations independently. By the correlation

theorem,

$$A \star B = \mathcal{F}^{-1} \left[\overline{\mathcal{F}[A]} \mathcal{F}[B] \right],$$

which allows efficient evaluation using FFT. We zero-pad the arrays to remove circular wrap-around [25] and cache the map-side spectra of V , F , and F^2 across headings.

To prevent weak-overlap candidates from receiving spuriously low costs, we discard translations satisfying

$$n_\psi(\tau) < \eta \sum_{\mathbf{u}} m_\psi(\mathbf{u}).$$

Among all remaining translation–heading pairs, we retain the K candidates with the lowest E_{ret} . We convert a retained lattice translation to a metric map position as

$$\mathbf{c} = \mathbf{o}_M + \Delta \tau.$$

E. Local Refinement

Retrieval is global but discrete; refinement is local but continuous. The retrieval stage quantizes translation and heading and scores a spline-derived raster, so we refine every retained candidate using measured LiDAR points directly against the continuous map field f_M . To limit the influence of the densely sampled near range, we draw a fixed subset of N_{ref} points using equal per-band quotas from concentric horizontal-range bands; the same subset is used for every candidate and iteration of a scan.

For point i at pose ξ , define the height residual and its mean as

$$r_i(\xi) = z_i - f_M(\mathbf{p}_i(\xi)), \quad \bar{r}(\xi) = \frac{1}{N_{\text{ref}}} \sum_i r_i. \quad (9)$$

The refinement cost is the same mean-centred height-error form used in retrieval, now evaluated on measured points:

$$E_{\text{ref}}^2(\xi) = \frac{1}{N_{\text{ref}}} \sum_{i=1}^{N_{\text{ref}}} (r_i - \bar{r})^2. \quad (10)$$

Because retrieval and refinement use different sampling—supported lattice cells versus measured points—their minimizers need not coincide, and we re-rank the candidates after refinement.

For evaluations in the map interior, differentiating through $\mathbf{p}_i(\xi)$ gives the 1×3 Jacobian

$$\mathbf{J}_i = -\nabla f_M(\mathbf{p}_i)^\top [\mathbf{I}_2 \quad \dot{\mathbf{R}}_\psi \mathbf{s}_i], \quad (11)$$

where $\dot{\mathbf{R}}_\psi = \partial \mathbf{R}_\psi / \partial \psi$. The centred residual $\tilde{r}_i = r_i - \bar{r}$ has Jacobian $\tilde{\mathbf{J}}_i = \mathbf{J}_i - \frac{1}{N_{\text{ref}}} \sum_j \mathbf{J}_j$. Stacking these into $\tilde{\mathbf{r}}$ and $\tilde{\mathbf{J}}$, each Gauss-Newton increment solves $\delta \xi \in \arg \min_{\delta} \|\tilde{\mathbf{r}} + \tilde{\mathbf{J}}\delta\|_2^2$. We apply fixed component-wise step limits before updating the pose, residuals, and Jacobians. After all K hypotheses are refined, we return the candidate pose with the lowest final E_{ref} ; the result need not lie on the retrieval lattice.

F. Implementation Details

Unless otherwise stated, we use $h_0 = h_S = \Delta = 4$ m, $\rho = 3$ m, $N_\psi = 24$, $\eta = 0.85$, and $K = 100$. For spline fitting, the nominal regularization parameters are $\lambda_M = \lambda_S = 10^{-3}$. Their effective penalty weights are scaled to the corresponding data and penalty matrices as

$$\alpha = \lambda \frac{\text{tr}(\mathbf{H})}{\max(\text{tr}(\mathbf{P}), 10^{-12})},$$

where $\mathbf{H}$ is the data Gram matrix and $\mathbf{P}$ is the second-difference penalty matrix. We apply this scaling separately to each DEM axis and to the two-dimensional scan fit. We median-fill missing DEM cells for fitting, but they remain invalid in V .

Refinement samples up to 80 points per horizontal-range band with boundaries $\{0, 5, 15, 30, 60, 250\}$ m and a fixed seed, giving $N_{\text{ref}} \leq 400$, and runs at most $I = 60$ iterations per candidate with component-wise step limits of 10 m in translation and 0.25 rad in heading, stopping when the optimization increment satisfies the predefined convergence thresholds.

IV. EVALUATION

A. Lunar Dataset and Evaluation Protocol

We use a lunar simulation dataset constructed with OmniLRS [27]. We initialize the four environments from real orbital topography: Apollo 11, Apollo 14, and Apollo 15 from 2 m/pixel LROC NAC DTMs [2], and Haworth from a 1 m/pixel shape-from-shading (SfS) DEM [3]. The reference maps span 300×320 m at Apollo 11, 2000×2000 m at

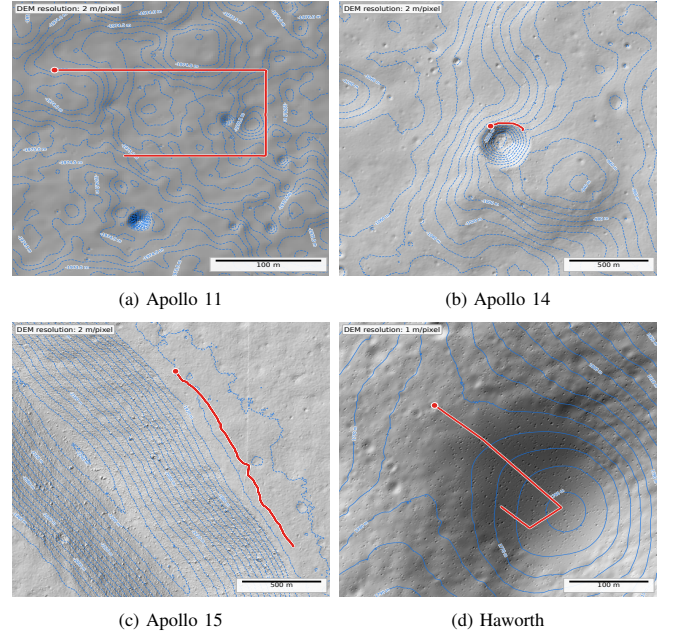


Fig. 5. **Lunar dataset sites and rover trajectories.** Selected lunar terrain regions used for LiDAR data acquisition. Red curves denote rover trajectories, blue curves denote elevation contours. Source DEM resolution and spatial scale are annotated in each panel.

Apollo 14 and Apollo 15, and 400×400 m at Haworth. The sites cover contrasting terrain: low-relief mare at Apollo 11, crater relief at Apollo 14, directional relief near Hadley Rille at Apollo 15, and extended crater-wall slopes at Haworth. Fig. 5 illustrates the four lunar sites and the corresponding rover trajectories used for LiDAR data acquisition.

The simulated rover carries a 128-channel LiDAR with a 360° horizontal field of view, corresponding to an Ouster OS1-128 class sensor, as hardware standard for NASA HI-RATE lunar project. We render scans instantaneously and subsample the scan sequence at 1 m intervals between acquisition positions. We select 100 queries per site and keep them fixed across all methods and resolutions. No odometry, temporal accumulation, position initialization, or heading initialization is used.

We export each simulated terrain at 0.25 m/pixel and area-mean downsample it to 0.5, 1, 2, 5, and 10 m/pixel reference maps. Thus, 0.25 m/pixel denotes the simulation-terrain sampling rather than an orbital observation; Sec. IV-E separately evaluates the native orbital products. We report median horizontal error $e_{xy} = \|\hat{\mathbf{c}} - \mathbf{c}\|_2$, success rate within 5 m (SR@5m), and, for SplineLoc, median heading error e_ψ . For DARCES, which may return no estimate, #Est. denotes the number of queries with a returned pose out of the 100 fixed queries at each site-resolution condition. We compute its median error only over returned estimates, while missing estimates count as failures in SR@5m. We use the same SplineLoc configuration at every resolution, with only $h_M = \max(4 \text{ m}, r_D)$ changing with the DEM.

B. Baselines

We compare SplineLoc with DARCES [6], BEV-Place++ [5], RING [4], and multi-start ICP [26]. We reim-

TABLE I

LOCALIZATION PERFORMANCE ACROSS DEM RESOLUTIONS. MEDIAN IS THE HORIZONTAL POSITION ERROR AND SR@5M THE SUCCESS RATE WITHIN 5 M. FOR DARCES, #EST. IS THE NUMBER OF RETURNED POSE ESTIMATES OUT OF 100 FIXED QUERIES PER SITE-RESOLUTION CONDITION; ITS MEDIAN IS COMPUTED OVER RETURNED ESTIMATES ONLY, WHILE MISSING ESTIMATES COUNT AS FAILURES IN SR@5M.

Site	DEM res. (m/pixel)	DARCES [6]			BEVPlace++ [5]		RING [4]		ICP [26]		SplineLoc (Ours)	
		Median (m)	SR@5m (%)	#Est.	Median (m)	SR@5m (%)	Median (m)	SR@5m (%)	Median (m)	SR@5m (%)	Median (m)	SR@5m (%)
Apollo 11	0.25	—	0.0	0/100	101.25	15.0	58.06	24.0	84.97	1.0	1.55	98.0
	0.5	5.63	1.0	3/100	101.86	15.0	50.81	29.0	83.30	1.0	1.55	98.0
	1	3.53	12.0	18/100	101.99	13.0	51.68	28.0	85.49	0.0	1.55	98.0
	2	5.49	12.0	26/100	102.14	10.0	59.71	26.0	85.59	0.0	1.54	98.0
	5	5.84	15.0	30/100	104.55	10.0	66.86	23.0	82.95	1.0	1.62	98.0
	10	40.54	0.0	9/100	102.98	1.0	65.75	5.0	95.88	1.0	1.86	96.0
Apollo 14	0.25	2.64	2.0	2/100	1.49	78.0	1.66	74.0	52.02	0.0	1.05	98.0
	0.5	2.35	37.0	38/100	1.40	84.0	1.48	77.0	52.03	0.0	1.05	98.0
	1	2.86	40.0	44/100	1.61	85.0	1.68	80.0	52.27	0.0	1.05	98.0
	2	3.54	22.0	31/100	2.33	85.0	2.42	79.0	52.24	1.0	1.05	98.0
	5	5.36	9.0	24/100	41.44	28.0	8.40	25.0	63.79	0.0	1.03	99.0
	10	—	0.0	0/100	220.79	0.0	171.82	1.0	91.44	2.0	1.22	96.0
Apollo 15	0.25	1.39	35.0	43/100	0.89	82.0	1.23	88.0	570.67	0.0	1.10	100.0
	0.5	1.52	71.0	84/100	0.90	83.0	1.23	90.0	557.28	0.0	1.10	100.0
	1	1.54	88.0	97/100	1.12	84.0	1.46	87.0	574.82	0.0	1.10	100.0
	2	1.99	80.0	94/100	2.06	71.0	1.78	91.0	612.56	0.0	1.10	100.0
	5	4.69	46.0	83/100	418.10	19.0	4.55	55.0	557.44	0.0	1.18	100.0
	10	318.65	0.0	58/100	548.23	0.0	458.35	2.0	535.28	0.0	1.84	97.0
Haworth	0.25	—	0.0	0/100	44.64	8.0	16.69	31.0	92.62	0.0	1.56	83.0
	0.5	12.88	4.0	19/100	59.24	7.0	22.18	28.0	92.64	0.0	1.56	82.0
	1	52.38	4.0	44/100	53.99	5.0	20.47	27.0	92.66	0.0	1.56	82.0
	2	145.35	1.0	41/100	58.84	2.0	28.10	22.0	90.43	0.0	1.56	81.0
	5	227.54	0.0	6/100	60.79	2.0	48.60	14.0	92.43	0.0	1.58	81.0
	10	—	0.0	0/100	63.98	3.0	50.82	3.0	87.41	0.0	1.79	78.0

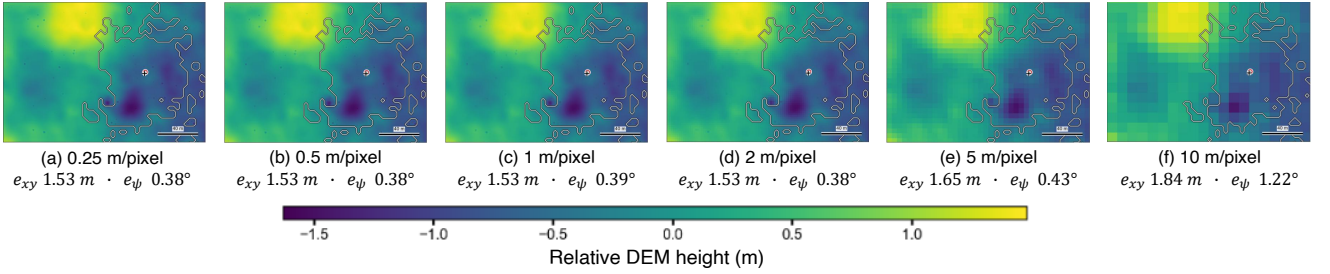


Fig. 6. **Qualitative localization across DEM resolutions.** SplineLoc localizes the same LiDAR query independently against reference DEMs from 0.25 to 10 m/pixel. The outlined region shows the LiDAR support at the estimated pose, while the markers denote the ground-truth and estimated positions. SplineLoc maintains a stable pose estimate as the DEM becomes progressively coarser.

plemented DARCES from the published formulation because no official code was available to us; the other baselines use official code. For each baseline, we test unlevelled scans and scans levelled with simulator-provided roll and pitch, report the better-performing variant. Among DARCES, DARCES followed by RANSAC, and the complete single-frame MOGA pipeline, DARCES achieves the best performance among these variants and is therefore reported.

We have fine-tuned BEVPlace++ on Apollo 14 LiDAR-DEM pairs with training runs separate from the held-out query run; the same checkpoint is used at all sites. RING retains its sinogram representation and the fixed $z > -0.6$ m filter on query and reference clouds. As a local-registration control, ICP uses ten uniformly sampled planar initializations with zero rotation and returns the highest-fitness result.

We evaluate BEVPlace++ and RING after their complete pipelines, including pose refinement. We render their reference views from the DEM at 5 m intervals using a virtual LiDAR (height 0.415 m, 128×2048 rays, first intersections), not a prior traverse. BEVPlace++ encodes 201×201 count-BEV images at 0.4 m/pixel with REIN into 8192-dimensional descriptors. At Apollo 14/15, the databases cover the query-trajectory bounding box expanded

by 40 m (BEVPlace++) or 80 m (RING); at Apollo 11 and Haworth, they cover the supplied DEM. SplineLoc searches the full supplied DEM subject to its overlap criterion, a larger domain at Apollo 14/15. Although BEVPlace++ and RING benefit from learned/database-based retrieval formulations, they are evaluated with complete pipelines including their respective retrieval and refinement stages. In contrast, SplineLoc does not use training data or pre-built reference scans and directly evaluates hypotheses against the supplied DEM.

C. Localization across Resolutions and Terrains

Table I shows that SplineLoc achieves the highest SR@5m in all 24 site-resolution conditions, with median horizontal errors of 1.03–1.86 m. Across the four sites, mean SR@5m decreases from 94.75% at 0.25 m/pixel to 91.75% at 10 m/pixel. The map/scan splines and retrieval lattice retain a 4 m nominal scale for 0.25–2 m/pixel. At 5 and 10 m/pixel, the map knot spacing follows the DEM resolution, while matching uses relative heights over the scan footprint.

This behavior contrasts with the reference-scan database baselines. At 5 and 10 m/pixel, SplineLoc achieves 94.50% and 91.75% SR@5m, whereas RING, the strongest baseline at these resolutions, achieves 29.25% and 2.75%. BEV-

TABLE II

EFFECT OF TERRAIN REPRESENTATION UNDER COARSE DEMs. RESULTS ARE POOLED OVER APOLLO 14 AND APOLLO 15 (200 QUERIES PER CONDITION). MED. AND e_ψ DENOTE THE FINAL MEDIAN HORIZONTAL (M) AND HEADING (DEG) ERRORS. $R@K$ DENOTES PRE-REFINEMENT RETRIEVAL RECALL: AT LEAST ONE OF THE K CANDIDATES IS WITHIN 5 M AND 15° OF GROUND TRUTH.

Representation	2 m/pixel				5 m/pixel				10 m/pixel				20 m/pixel				50 m/pixel			
	Med.	SR@5m	e_ψ	$R@K$	Med.	SR@5m	e_ψ	$R@K$	Med.	SR@5m	e_ψ	$R@K$	Med.	SR@5m	e_ψ	$R@K$	Med.	SR@5m	e_ψ	$R@K$
B-spline (Ours)	1.06	99.0	0.20	96.5	1.13	99.5	0.24	96.5	1.51	96.5	0.39	96.5	2.34	83.5	0.63	95.5	11.83	21.5	3.79	82.0
Bilinear	1.05	99.5	0.20	96.5	1.17	99.5	0.29	96.5	1.50	95.5	0.53	96.5	3.03	75.5	1.12	95.0	28.85	0.5	11.33	75.5
NN interpolation	1.08	99.0	0.25	96.5	1.23	99.0	0.31	96.5	2.07	91.0	0.69	96.5	3.97	56.0	1.14	95.5	817.57	0.5	55.63	37.5

Place++ and RING remain competitive on Apollo 14 and Apollo 15 at finer resolutions, but their performance degrades sharply as we coarsen the reference map. The results therefore suggest that SplineLoc relies less on fine-scale local geometry and instead matches broader height variation preserved across DEM resolutions.

DARCES illustrates the distinction between estimate availability and accuracy (Table I). At Apollo 14 and 0.25 m/pixel, its 2.64 m median comes from only two returned estimates out of 100 queries, with 2% SR@5m. At Apollo 15, it achieves 88% SR@5m at 1 m/pixel, but returns 58 estimates with no successes at 10 m/pixel. Thus, failures include both missing estimates and returned poses outside the success threshold. We therefore interpret the median over returned estimates alongside SR@5m and #Est., rather than as a standalone measure of localization reliability.

Apollo 14 and Apollo 15 provide the main evidence for map-wide retrieval because both reference maps span 2000×2000 m. At 10 m/pixel, SplineLoc still achieves 96% and 97% SR@5m with median errors of 1.22 and 1.84 m, respectively. The final estimates need not lie on DEM cells or the retrieval lattice because refinement evaluates the continuous map field directly. Thus, sub-cell position accuracy reflects continuous alignment of terrain structure over the scan footprint rather than recovery of terrain detail absent from the coarse DEM.

Apollo 11 represents a different regime: its 300×320 m map is comparable to the scan footprint and contains weak low-relief structure. Nevertheless, SplineLoc maintains 96–98% SR@5m with median errors below 2 m, indicating that a prominent extracted landmark is not required when sufficient distributed height variation is available.

Haworth remains the most challenging site, with SR@5m decreasing from 83% to 78% across the resolution sweep. Its extended crater-wall segments contain similar local surface normals. On a locally planar region, translation primarily introduces an approximately constant height change, which the offset-invariant objective removes. Large slope magnitude alone is therefore not necessarily distinctive; spatial variation in terrain gradient is more informative for resolving position. **Qualitative localization.** Fig. 6 tracks the same LiDAR query across all six DEM resolutions. The estimated pose remains nearly unchanged from 0.25 to 2 m/pixel, with a horizontal error of about 1.53 m, and degrades only moderately as the DEM becomes coarser. Even at 10 m/pixel, where individual DEM cells are clearly visible, SplineLoc localizes the query with a 1.84 m horizontal error and a

TABLE III

REFERENCE MAPS FROM NATIVE ORBITAL PRODUCTS. THE MAP CHANGES TO THE LROC NAC DTM OR SFS DEM AND ITS AREA-MEAN DOWNSAMPLING TO 5 AND 10 M/PIXEL, AFTER APPLYING A METADATA-BASED SITE-WISE TRANSLATION SHARED BY ALL QUERIES.

Site	r_D	Sim.-derived		Native product		
		Med.	SR@5m	Med.	SR@5m	e_ψ
Apollo 11	2	1.54	98.0	1.54	98.0	0.70
	5	1.62	98.0	1.64	98.0	0.70
	10	1.86	96.0	1.87	95.0	1.05
Apollo 14	2	1.05	98.0	1.05	98.0	0.19
	5	1.03	99.0	1.04	98.0	0.23
	10	1.22	96.0	1.21	97.0	0.31
Apollo 15	2	1.10	100.0	1.08	100.0	0.23
	5	1.18	100.0	1.22	100.0	0.28
	10	1.84	97.0	1.80	98.0	0.59
Haworth	1	1.56	82.0	1.57	82.0	0.59
	5	1.58	81.0	1.59	79.0	0.69
	10	1.79	78.0	1.80	78.0	0.77

1.22° heading error. This example is consistent with the quantitative results, showing that the localization remains stable despite substantial DEM coarsening.

D. Effect of Terrain Representation

Table II studies the impact of terrain representation under coarse DEMs. We compare the proposed B-spline field against nearest-neighbor and bilinear interpolation while keeping the localization pipeline unchanged. Results are evaluated on Apollo 14 and Apollo 15 due to their large-scale map-wide localization setting.

All representations perform similarly at fine resolutions, where the reference DEM preserves sufficient terrain structure. As the DEM becomes coarser, however, discrete interpolation methods exhibit increasing degradation. NN interpolation is particularly sensitive to height quantization, and bilinear interpolation also deteriorates at large cell sizes.

In contrast, the proposed B-spline representation maintains a more gradual performance degradation as the reference DEM becomes increasingly coarse. At 50 m/pixel, where individual DEM cells no longer capture meaningful local geometry, SplineLoc still achieves 21.5% SR@5m with 11.83 m median position error and 3.79° heading error. In comparison, bilinear interpolation drops to 0.5% SR@5m with 28.85 m median error and 11.33° heading error, while NN interpolation completely fails with 817.57 m median

error and 55.63° heading error.

This result indicates that the benefit of the continuous representation becomes increasingly important when the reference map resolution approaches the scale at which discrete cells cannot reliably describe terrain variation.

E. Reference Maps from Native Orbital Products

The resolution sweep uses maps exported from the query-generating terrain, including procedural craters. We replace them with native orbital products (Apollo: 2 m/pixel LROC NAC DTMs; Haworth: 1 m/pixel Sfs DEM) and their area-mean downsampling to 5 and 10 m/pixel, keeping queries and configuration fixed. We crop and transform each native map to the simulation frame by a site-wise horizontal translation derived solely from coordinate and crop metadata, not query scans or their ground-truth poses. The same translation applies to all queries and resolutions.

SR@5m changes by at most two percentage points at every site and resolution (Table III). Both map variants share the underlying orbital terrain; only the simulation-derived maps contain the added craters. The similar performance suggests that the shared terrain structure remains informative despite these differences.

F. Computational Cost

We average timings over eight queries per site in an isolated single-process CPU run (Intel Core i5-13400F, 64 GB DDR5-5600). Offline map fitting takes at most 0.22 s (2.5×10^5 coefficients at Apollo 14/15, 2 m/pixel). Online runtimes are 1.6 s at Apollo 11, 4.5–5.1 s at Apollo 14/15, and 5.7–5.9 s at Haworth, with little variation across DEM resolutions. Scan fitting dominates (0.7–4.9 s, increasing with point count); FFT retrieval takes 0.15–1.24 s depending on map area, and refinement takes 0.5–1.3 s. Database baselines also require pseudo-LiDAR reference rendering: about 2 h per resolution for trajectory-restricted Apollo 15. SplineLoc targets periodic global fixes rather than per-frame estimation, allowing stationary scan acquisition.

V. CONCLUSION

We have presented SplineLoc, a single-scan LiDAR-to-DEM localization method that represents both observations as continuous cubic B-spline height fields. SplineLoc performs global retrieval using an offset-invariant height-matching objective and masked FFT, followed by continuous refinement and re-ranking on measured LiDAR points. The method requires no position or heading initialization, terrain-landmark extraction, or a reference-scan database.

Experiments across multiple lunar terrains and DEM resolutions show that SplineLoc consistently outperforms the evaluated baselines and maintains high localization accuracy as the reference DEM becomes progressively coarser. Similar performance with native orbital products further suggests that the method exploits terrain structure preserved across map representations rather than simulation-specific fine-scale detail. Future work will address ambiguous terrain with weak spatial variation in surface geometry and evaluate robustness to realistic attitude and sensor uncertainties.

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